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Plant-Scale Biogas Production Based on Integrating of CEEMDAN Decomposition with PSO Optimized Multilayer Perceptron Neural Network

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Abstract: Accurate and dependable forecasting of biogas production is vital for optimizing process parameters and maintaining stable operation in large-scale anaerobic digestion projects. In this study, a novel hybrid approach (CEE-PMLP) integrating complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) and a multilayer perceptron (MLP) neural network optimized by particle swarm optimization (PSO) were proposed for predicting biogas production in large-scale anaerobic digesters (ADs). The methodology involves extracting Intrinsic Mode Function (IMF) components using CEEMDAN and subsequently employing MLP optimized by particle swarm optimization (PSO) to predict each component. The performance of the models was evaluated using root mean square error (RMSE), mean squared error (MSE), mean absolute error (MAE), and fitting determination coefficient (R^2). The findings revealed that the prediction errors of the proposed CEE-PMLP model were consistently lower than those of other comparative models. Notably, the model achieved the highest R^2 value of 98%, indicating an exceptionally high accuracy in prediction. The validation experiment confirmed the high accuracy of the CEE-PMLP model, further demonstrating its superiority in biogas production prediction.

Keywords: biogas production prediction; complete ensemble empirical mode decomposition with adaptive noise; multilayer perceptron neural network; particle swarm optimization; large-scale anaerobic digestion project



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1. Introduction

Anaerobic digestion (AD) is a widely adopted technology for transforming organic waste into valuable bioenergy (methane-rich biogas, CO₂ and nutrient replacements) [1,2]. This process is recognized as a promising renewable and sustainable energy production technology that not only enables energy recovery, but also helps to reduce environmental pollution from organic waste [3]. Despite its extensive use in organic waste disposal, adjusting operational parameters to optimize the performance and efficiency of AD remains challenging due to the complexity of the microbial communities, fermentation processes, and feed streams involved [4,5]. Therefore, it is important to properly regulate the operating parameters and take necessary precautions to maintain the stability and efficiency of the fermentation process [6].

Mathematical models play a crucial role in enhancing and predicting the fermentation of organic matters in AD. These models are used to simulate and optimize biogas production efficiency [7]. They involve a detailed analysis of the metabolic pathways and

physicochemical processes that occur during fermentation, typically represented through mathematical equations [8,9]. Although these models offer a high degree of interpretability, their development is challenging and requires an in-depth understanding of the fermentation system [10]. Recently, with the development of computational algorithms and the improvement in the accessibility of computing power, machine learning has attracted significant attention as a promising approach to data mining and model development [11]. Machine learning can provide insights into potentially complex interactions between numerous operating parameters and output outcomes to enable output predictions [12]. There are many successful practices in predicting methane production from various feedstocks through machine learning. Xu et al. predicted methane production from lignocellulosic waste during 30-day AD using an artificial neural network model [13]. Wang et al. used a trained K-nearest neighbor model to accurately predict methane production from the anaerobic digestion of lignocellulosic biomass [14]. Multilayer perceptron (MLP) is one of the most popular and widely used methods in machine learning, serving as a feed-forward neural network designed to process and analyze complex data patterns [15,16]. However, when it comes to predicting biogas production, constructing a single robust machine learning model presents significant challenges [17]. This is due to the multi-stage nature of anaerobic digestion, which includes hydrolysis, acidogenesis, acetogenesis, and methanogenesis, making the process highly nonlinear and intricate. To overcome the limitations of single machine learning models, hybridization based on preprocessed signal decomposition has recently become popular [18].

Complete ensemble empirical-mode decomposition with adaptive noise (CEEMDAN) is an advanced signal-processing technique designed to decompose nonlinear and nonstationary signals into a set of Intrinsic Mode Functions (IMFs) [19]. These IMFs capture the distinct frequency components inherent in the original signal, facilitating the extraction of significant patterns for in-depth analysis. They provide a decomposition that better separates the frequency components, allowing for more effective modeling of nonstationary processes [20]. The integration of CEEMDAN with machine learning models for biogas prediction may offer a promising solution to the challenges posed by data non-stationarity and fluctuations within the AD process.

Therefore, a novel approach that improves the accuracy and reliability of biogas prediction by combining advanced signal processing techniques and machine learning algorithm was proposed in this study. This process began with the application of CEEMDAN as a preprocessing step to decompose complex, nonlinear biogas production data into multiple IMFs, capturing different frequency patterns that correspond to different aspects of the AD process. Concurrently, the multilayer perceptron (MLP) neural network was used to predict biogas production, while the Particle Swarm Optimization (PSO) was employed to optimize the neural network's structural parameters, such as the number of neurons, layers, and learning rate. Subsequently, features extracted via CEEMDAN were used as inputs for predicting biogas production within the optimized MLP. The network parameters were adjusted by minimizing the error function, and the predicted values from each mode component were aggregated to derive the final prediction result. This hybrid approach is expected to outperform traditional methods by leveraging both advanced signal processing and machine learning techniques, offering valuable insights for biogas production, environmental monitoring, and sustainable energy management.

2. Materials and Methods

2.1. Basic Information of the Dataset

In this study, daily operational data of the biogas production system from 5 September 2022 to 31 March 2023 was gathered from an anaerobic agricultural waste treatment plant (Hebei, China). This plant utilizes a continuous stirred-tank reactor for processing solid waste, including straw and manure. The system comprises four adjacent reactors, each with an effective volume of 5262 m³. Key operational parameters including dry matter (DM), organic matter (OGM), total alkalinity (TAC), free organic acids (FOS), FOS/TAC

ratio, temperature (T), dissolved oxygen (DO), and pH were selected as input variables for developing the predictive model. Biogas production was used as the output variable to evaluate the model's accuracy. To evaluate the model's performance, the last 30 data points were used as the test set and the other data served as the training set.

2.2. Data Processing Through CEEMDAN

CEEMDAN is a cutting-edge signal-processing technique designed to decompose nonlinear and nonstationary data into multiple intrinsic mode functions (IMFs), which represent distinct frequency components of the original signal. By introducing adaptive noise, CEEMDAN enhances the precision and stability of signal decomposition, addressing mode-mixing issues prevalent in empirical mode decomposition (EMD) and improving upon ensemble empirical mode decomposition (EEMD). This makes CEEMDAN highly effective for analyzing complex signals and provides a robust basis for integrating with machine learning models in applications such as biogas prediction.

CEEMDAN is a signal-processing technique that breaks down nonlinear and non-stationary data into multiple intrinsic mode functions (IMFs), resulting in a more stable and stationary series of IMFs [21]. CEEMDAN adds extra features or improvements to the original empirical mode decomposition (EMD) algorithm, providing a more comprehensive signal analysis capability [22]. EMD, first introduced by Huang et al. [23], suffers from mode-mixing issues, which prompted the creation of ensemble empirical mode decomposition (EEMD). EEMD resolves mode mixing more effectively than EMD by introducing noise-assisted decomposition [24]. However, EEMD still has drawbacks, such as high computational cost and subjective parameter settings [25]. CEEMDAN further enhances precision and robustness by using adaptive noise instead of the fixed noise in EEMD, achieving better reductions in mode mixing and more accurate signal decomposition and reconstruction [26]. The CEEMDAN procedure consists of the following steps:

Adding white noise: To incorporate white noise into the original signal $X(t)$, a sequence of signals is generated by adding white noise $u_n(t)$ with a standard deviation of ε , resulting in a new sequence of signals $X_n(t) = X(t) + \varepsilon u_n(t)$, where $n = 1, 2, \dots, K$, K denotes the number of implementations.

EMD decomposition: Empirical mode decomposition (EMD) is applied to each signal realization to determine the first IMF.

$$\text{Imf}_1(t) = \frac{1}{K} \sum_{i=1}^K \text{IMF}_1(t) \quad (1)$$

Calculating the residual: The residual from the first stage is calculated as $r_1(t) = X(t) - \text{Imf}_1(t)$; Then, $r_1(t) + \varepsilon_1 \text{EMD}_1(u_n(t))$ is further decomposed through EMD to calculate the second IMF and its corresponding residual.

Decomposition at the k -th stage: At the k -th stage, the k -th residual and the $(k + 1)$ -th IMF are calculated using the following formula:

$$r_k(t) = r_{k-1}(t) - \text{Imf}_k(t) \quad (2)$$

$$\text{Imf}_{k+1}(t) = \frac{1}{K} \sum_{n=1}^K \text{EMD}_1(r_k(t) + \varepsilon_k \text{EMD}_k(u_n(t))) \quad (3)$$

These steps are repeated until the residuals can no longer be decomposed; eventually, the original signal is broken down into:

$$X(t) = \sum_{i=1}^N \text{Imf}_i(t) + R(t) \quad (4)$$

where N represents the number of samples, and $R(t)$ is the final residual component acting as the residual term.

In the end, the original signal is broken down into multiple IMFs along with a final residual component, representing the residual term. This structured decomposition allows for a detailed analysis of the data's inherent patterns and fluctuations across various scales.

2.3. Optimization of MLP Model by PSO Algorithm

The MLP is a type of feedforward neural network consisting of multiple neurons, typically organized into an input layer, one or more hidden layers, and an output layer [27]. The MLP structure and hyperparameters significantly influence its capacity to model complex nonlinear relationships. Therefore, selecting the right parameters is essential for ensuring the effectiveness and accuracy of MLP networks [28]. To enhance the prediction accuracy of MLP, the Particle Swarm Optimization (PSO) algorithm was employed to optimize key hyperparameters, including the learning rate, the number of hidden layers, and the number of neurons per layer. PSO is inspired by the simulation of the social behavior of birds in a flock and is effective in solving search and optimization problems [29,30]. In order to maximize prediction accuracy, PSO was employed to optimize the structural parameters of the MLP, including the learning rate, number of network layers, and number of neurons per layer. As shown in Figure 1, the optimization steps are as follows.

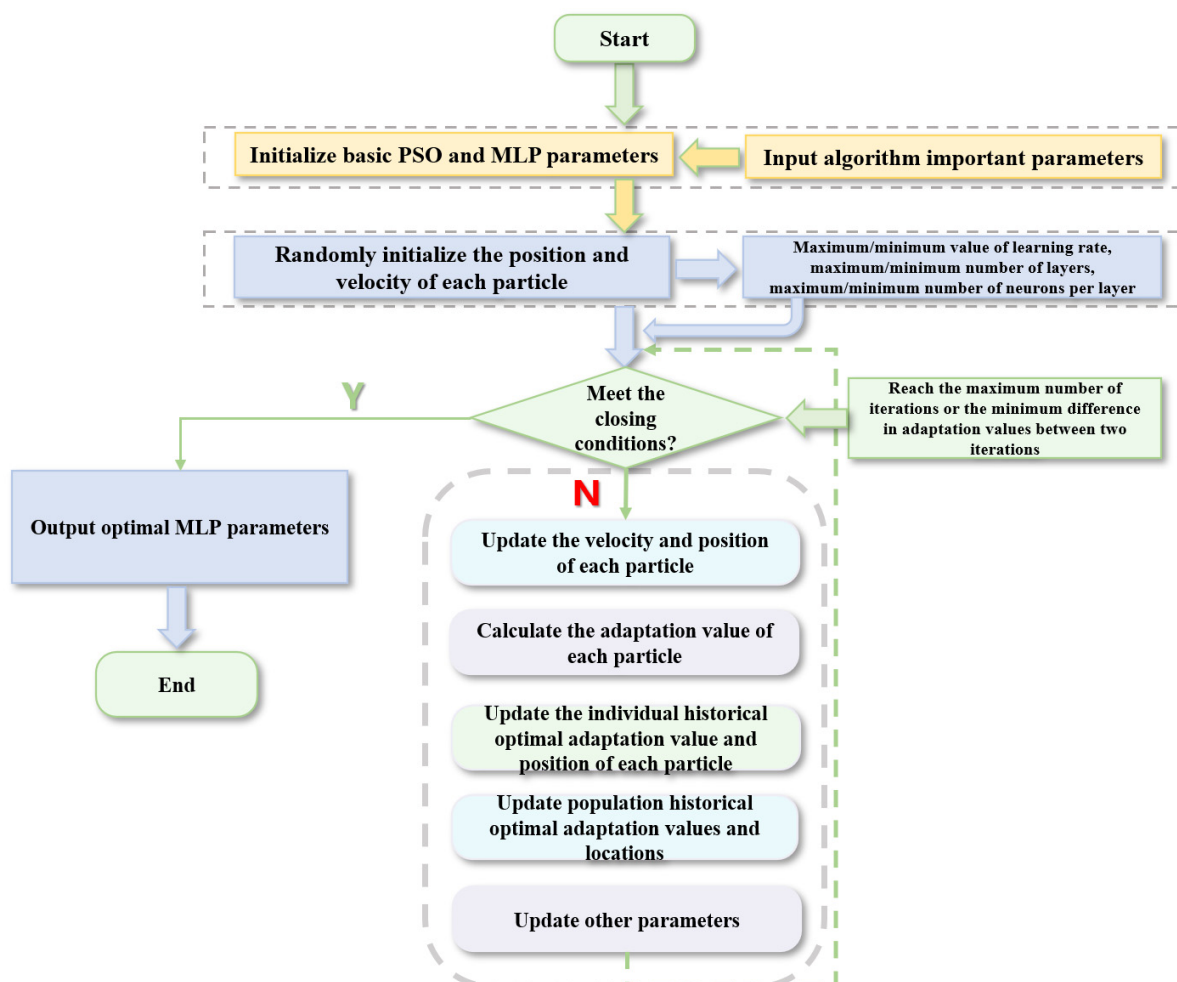


Figure 1. Flowchart of the MLP optimized by the PSO algorithm.

Additionally, the hyperparameter ranges and selection rationale are crucial for tuning the MLP's performance.

Learning Rate (0.00001–0.05):

The learning rate controls how much the model updates its weights with each iteration. A smaller learning rate ensures convergence but might slow down training, whereas a larger learning rate speeds up training, but risks instability. The range of 0.00001–0.05 was chosen based on preliminary experiments and literature recommendations, striking a balance between convergence stability and computational efficiency.

Number of Hidden Layers (2–8):

A deeper network can capture more complex patterns, but risks overfitting with insufficient data. For this study, 2 to 8 layers were tested, as this range is known to adequately represent nonlinear dynamics without excessive model complexity.

Number of Neurons per Layer (5–100):

More neurons enhance the network's representational capacity, but increase the risk of overfitting. The range of 5–100 neurons per layer was set to provide flexibility while keeping the computational cost manageable.

Dropout Rate (0.2):

Dropout was applied to mitigate overfitting by randomly deactivating neurons during training. A fixed rate of 0.2 was adopted, as it proved effective in balancing model performance and generalization.

PSO Parameters:

The number of particles was set to 25, sufficient for exploring the hyperparameter space while maintaining computational efficiency.

The inertia weight was set at 0.7, balancing exploration and exploitation in the search process.

Personal and global learning factors were set at 1.5 to ensure effective local and global search capabilities.

Optimization Process**Initialization:**

The search space was defined for each hyperparameter as specified in Table 1. Each particle was randomly initialized with a position and velocity within the defined bounds.

Table 1. MLP and initialized PSO parameters.

Parameters	Range/Value
MLP parameters	
Learning rate	0.0001–0.05
Max layers	8
Min layers	2
Max neurons per layers	100
Min neurons per layers	5
PSO parameters	
Numbers of particles	25
Max velocity	1.0
Min velocity	−1.0
Max position	1.0
Min position	−1.0
Inertia weight	0.7
Personal learning factor	1.5
Global learning factor	1.5

1. Fitness Evaluation:

The fitness function was defined as the mean absolute error (MAE) on a validation set (10% of training data). Each particle's fitness was calculated by training the MLP with its current hyperparameters.

2. Velocity and Position Update:

Using Equations (5) and (6), particle velocities and positions were updated based on their personal best positions and the global best position.

3. Convergence Criteria:

The PSO algorithm iteratively refined the particles' positions over 50 generations. The process terminated when the maximum number of generations was reached or no significant improvement in the global best fitness was observed.

Outcome of PSO Optimization:

The optimal configuration of the MLP hyperparameters obtained through PSO is presented in Table 1. This configuration was used for subsequent model training and prediction. The PSO-optimized MLP demonstrated superior predictive performance, with improved convergence stability and generalization ability compared to manually tuned hyperparameters.

Initialization: Define the parameter ranges for the MLP and initialize the parameters for PSO as specified in Table 1.

Evaluate initial solutions: For each particle, train the MLP with its existing parameters and test the fitness function on a validation set made up of 10% of the training data selected randomly.

Update velocity and position: Referencing Figure 2, for each particle, compare its current fitness function value with its personal best and the global best. If the current fitness function value is better than the personal best, update the particle's best position and value. If the current fitness function value is better than the global best, update the global best position and value accordingly. Use the following Formulas (5) and (6) to update the velocity and position of particles:

$$v_i^{(t+1)} = wv_i^{(t)} + c_1r_1(pbest_i - x_i^{(t)}) + c_2r_2(gbest - x_i^{(t)}) \quad (5)$$

$$x_i^{(t+1)} = x_i^{(t)} + v_i^{(t+1)} \quad (6)$$

where $v_i^{(t)}$ is the velocity of particle i at time t , w is the inertia weight (set to 0.7 here), c_1 and c_2 are the learning factors (set to 1.5 and 2.0, respectively), r_1 and r_2 are random numbers in the range of $[0, 1]$, $pbest_i$ is the individual best position of particle i , and $gbest$ is the global best position.

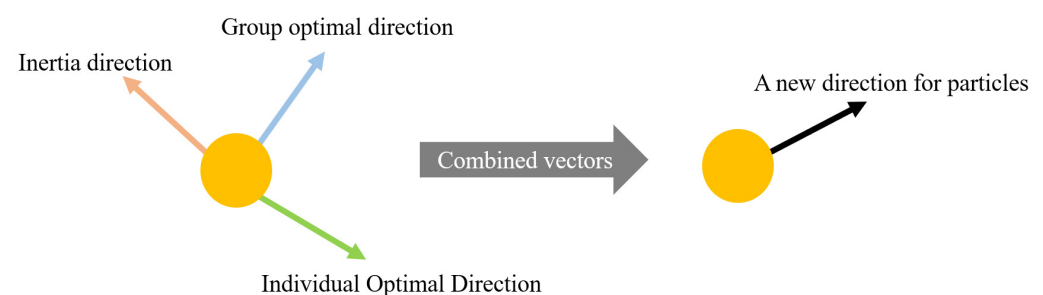


Figure 2. Schematic diagram of updated velocity and position.

Evaluate new solutions and update individual and global best: Use new MLP parameters, retrain the neural network, and calculate the fitness function value. Compare the new fitness function values with the individual and global best again and update accordingly (Figure 3). When the maximum number of iterations is reached, terminate the algorithm, return to the global best position, and define the optimal configuration of MLP.

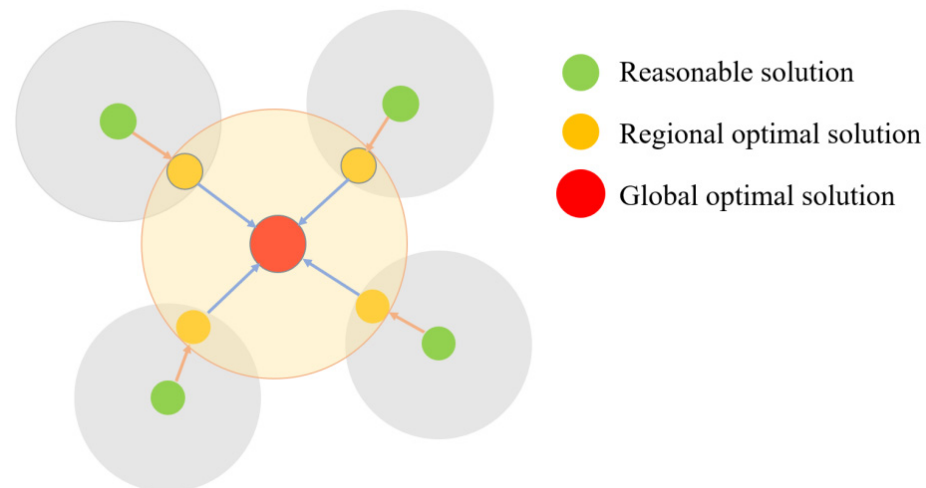


Figure 3. Schematic diagram of finding the local global optimal solution.

2.4. Proposed CEE-PMLP Prediction Model

By utilizing the advantage of CEEMDAN in extracting representative features from nonlinear and non-stationary data, as well as the advantage of MLP in handling nonlinear relationships, a hybrid approach for biogas prediction was developed in this study by combining CEEMDAN with PSO-optimized MLP (PMLP). The detailed steps of this method are as follows: Initially, after preprocessing, the biogas data were decomposed into smoother subseries (IMF1 through IMF6) using CEEMDAN. Subsequently, PSO was employed to optimize the hyperparameters of the MLP, refining the model through continuous training. Finally, the stabilized signal served as input to the PSO-optimized MLP network (PMLP) for biogas prediction.

In the model development process, inputs include obtained IMFs and eight fermentation parameters of DM, OGM, TAC, FOS, FOS/TAC, T, DO, and pH. The output of the model is as follows:

$$y_{\text{Biogas}} = \text{MLP} \left(\text{IMF1}, \text{IMF2} \dots, \text{IMF5}, \text{IMF6}, \text{Res}, \text{DM}, \text{OGM}, \text{TAC}, \text{FOS}, \frac{\text{FOS}}{\text{TAC}}, T, \text{DO}, \text{pH} \right) \quad (7)$$

When constructing the MLP, three neural network layers were employed with neuron counts of 65, 18, and 16, respectively. The initial learning rate was set to the default value of 0.011040419821353918. This specific value was generated through PSO during the optimization process and represents the globally optimal parameter for balancing convergence speed and model performance. The extended precision of this value ensures reproducibility and avoids rounding errors during iterative computations, which can affect the learning process, especially in deep learning models. The optimizer used was Adam, which combines the advantages of AdaGrad and RMSProp optimization algorithms, facilitating a natural annealing process and high computational efficiency. The loss function adopted was “MAE” (Mean Absolute Error), and a dropout rate of 0.2 was applied. The MLP’s hyperparameters were subsequently optimized using PSO, after which iterative training was utilized to enhance the model.

The CEE-PMLP model outperformed the EMD-PSO-MLP model, showing significant reductions in MAPE (17.5%), RMSE, and MAE and an R^2 value increase from 0.92 to 0.98. The comparison involved four single models, including four single-prediction models: GRU, MLP, RF, and SVR.

Two hybrid models, PSO-MLP and EMD-PSO-MLP, were also involved. In the validation process, these models were tested for biogas production prediction. MLP and GRU had 50 neurons each, and all models, including CEE-PMLP, used the Adam optimizer for 200 epochs with a convergence callback after 10 rounds. To mitigate the effects of randomness on the results, each test set was subjected to multiple repeated trials. The reported experimental results are averages of the evaluation metrics obtained from these repeated trials.

To better evaluate prediction accuracy, RMSE, MAPE, MAE, and R^2 were used to evaluate errors. The formulas are as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \quad (8)$$

$$\text{MAPE} = \frac{1}{n} \sum_{t=1}^n \left| \frac{\hat{y}_t - y_t}{y_t} \times 100\% \right| \quad (9)$$

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \quad (10)$$

$$R^2 = \frac{\sum_{t=1}^n (\hat{y}_t - \bar{y}_t)^2}{\sum_{t=1}^n (y_t - \bar{y}_t)^2} \quad (11)$$

where y_t represents the true values of the data from the last 30 test points and $\hat{y}_t$ represents the predicted values of the model.

The optimized hyperparameters from PSO were directly applied to the CEE-PMLP model. By incorporating CEEMDAN-decomposed IMFs as inputs, the MLP leveraged the refined signal characteristics, further enhancing the prediction accuracy.

3. Results and Discussion

3.1. Biogas Decomposition with CEEMDAN

In this study, datasets of biogas production under different operating parameters were collected. It is easy to see that there were sharp fluctuations in biogas production. Therefore, CEEMDAN was employed as a preprocessing technique to decompose the series data of biogas production, aiming to reduce the volatility of the data and improve the prediction accuracy (Figure 4). It can be seen that biogas production data were completely decomposed into six intrinsic mode function (IMF) components. In general, the IMF components are simpler compared with the original data of biogas production. The distribution of the IMF components presents lower complexity with fewer extreme values, which leads to more stable inputs for the subsequent model [31]. Analysis of the individual IMF showed that the fluctuation frequency gradually decreased from IMF1 to IMF6, with a notable reduction in volatility. The periodic fluctuations were particularly pronounced in the initial IMF1 to IMF3, aligning with the fluctuation patterns observed in the original data. Further investigation revealed that IMF1 to IMF3 predominantly captured mid-to-high-frequency fluctuations, whereas IMF4 and IMF5 mainly reflected the low-frequency trends in the data. Moreover, IMF6 served as the residual component, characterized by relatively small amplitude fluctuations. This indicates that the CEEMDAN decomposition effectively and thoroughly addressed the high-frequency components. Consequently, the periodic IMFs extracted from the biogas production data by CEEMDAN processing facilitated the subsequent MLP modeling, enhancing the model's predictive accuracy and reliability.

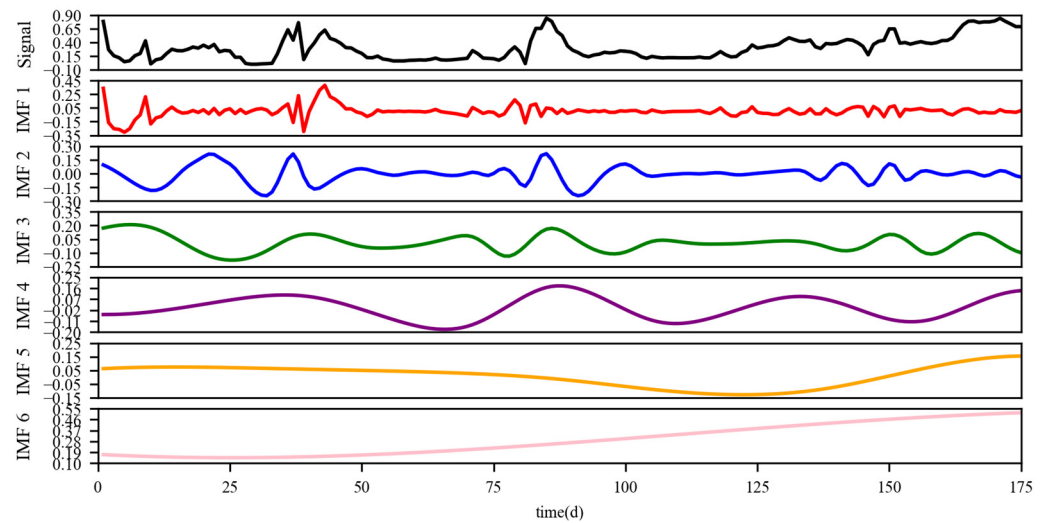


Figure 4. The variations in original biogas production and IMF components obtained by decomposing biogas production through CEEMDAN.

3.2. Assimilating Data Correlated with Biogas

To explore what was hidden in the decomposition variables of biogas production, Spearman correlation analysis was conducted on biogas and its decomposition variables. Table 2 illustrates that there were significant differences in the influencing factors of each IMF component after decomposition, indicating that the impact of these factors on biogas production is multi-scale rather than single-scale. Thus, it is essential to investigate the intrinsic connections between each factor and biogas production.

Table 2. Correlation analysis of influencing factors with biogas and decomposition variables.

Component Model	DM	OGM	T	DO	pH	FOS	TAC	FOS/TAC
Biogas	0.2 **	−0.216 **	−0.511 ***	−0.534 ***	0.74 ***	0.317 ***	0.289 ***	0.283 ***
IMF1	−0.021	0.071	0.003	−0.078	0.262 ***	0.078	−0.025	0.081
IMF2	0.181 *	0.067	−0.002	0.028	0.059	0.25 **	−0.057	0.243 **
IMF3	0.09	0.109	−0.26 ***	−0.483 ***	0.561 ***	0.14	−0.039	0.144
IMF4	0.004	−0.207 **	0.163	−0.137	0.069	0.014	−0.452 ***	0.093
IMF5	0.237 **	−0.296 ***	−0.594 ***	−0.318 ***	0.477 ***	0.193 *	0.543 ***	0.126
IMF6	0.255 **	−0.314 ***	−0.616 ***	−0.32 ***	0.474 ***	0.154	0.568 ***	0.083

DM, dry matter; OGM, organic matter; TAC, total alkalinity, FOS, volatile organic acids, T, temperature, DO, dissolved oxygen; ***, **, and *, respectively, indicate significance levels of 1%, 5%, and 10%.

As shown in Table 2, there was a significant positive correlation between biogas and DM, indicating that an increase in dry matter could boost biogas production by providing additional carbon sources and enhancing microbial activity. In contrast, there was a significant negative correlation with the quality of OGM, suggesting that biogas production may decline with the presence of hard-to-decompose organic materials. Furthermore, an increase in temperature correlated significantly with reduced biogas production, possibly because high temperatures affected the activity of key microorganisms or caused enzyme inactivation. Meanwhile, biogas production was positively correlated with pH, highlighting the importance of an appropriate pH level for maintaining microbial activity. Additionally, the biogas was positively correlated with FOS and TAC, indicating that the degree of organic matter decomposition and the amount of carbon available to microbes are crucial factors influencing biogas production. Notably, the ratio of FOS to TAC was also positively correlated with biogas production, implying that an increase in the proportion of free organic acids to total inorganic carbon boosted biogas generation. Interestingly, each operating parameter was significantly correlated with at least two IMF components,

suggesting that IMF components can better reflect the effects of operating parameters on biogas production. Furthermore, to validate the effectiveness of the CEEMDAN method in reducing data roughness, a new biogas sequence was obtained by fidelity reconstruction using six IMF component variables derived from biogas decomposition. As shown in Figure 5, the distributions of reconstructed biogas production data were extremely similar to those of the original data. This indicates that the CEEMDAN decomposition effectively preserved the integrity of the original data, ensuring the accuracy of data prediction.

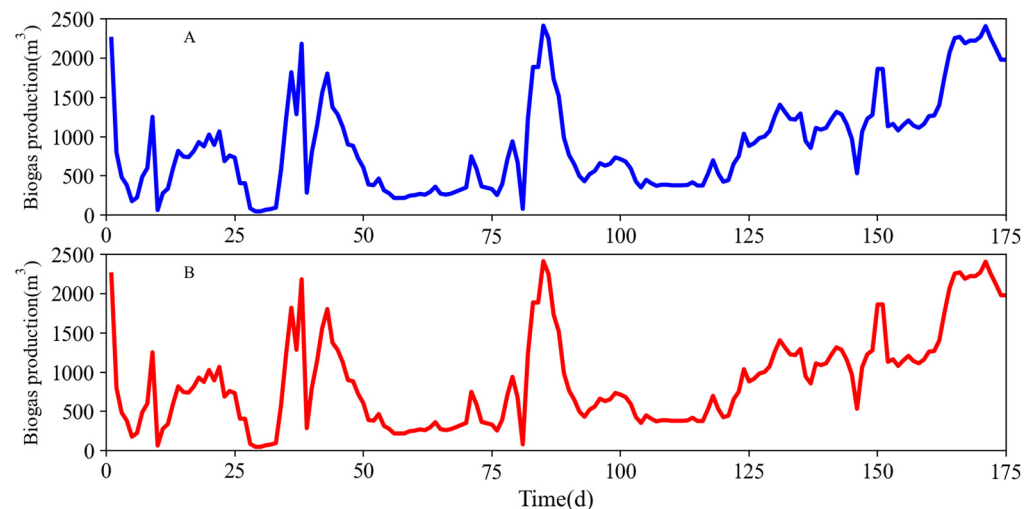


Figure 5. The distribution of biogas production during fermentation. (A) original data of biogas production; (B) reconstructed data of biogas production using six IMF components.

3.3. Optimization of MLP Model Structure Based on PSO

In order to reveal the optimization capability of the PSO algorithm in MLP, we compared the efficiency of the PSO algorithm with the Genetic Algorithm (GA). For the MLP model, the number of layers was defined between 2 and 4, as too many layers might lead to overfitting. Similarly, to prevent overfitting, the number of neurons was set between 5 and 100, and the learning rate was set between 0.00001 and 0.05 (Table 1). For the PSO algorithm, the initial particle population and the maximum number of iterations were set to 25 and 50, respectively. The inertia weight, individual learning factor, and global learning factor were set to the corresponding values (see Table 1). For the GA, we initialized a population size of 25, employed uniform crossover and integer range mutation, and set the number of generations to 50, aiming to find the optimal MLP parameters, which were within the same range as those for PSO. After testing, the total running time for the PSO algorithm was 80.97 s, while for the GA algorithm, it was 164.95 s, indicating the superiority of PSO in global optimization. Thus, optimizing the MLP model with the PSO algorithm led to the creation of the PMLP model. After data preprocessing, PMLP was used to forecast all sequences, and the results were aggregated to provide the final prediction.

3.4. Performance Evaluation of CEE-PMLP Model

The integration of CEEMDAN with the optimized PMLP formed the CEE-PMLP model proposed in this paper. As shown in Figure 6, high predictability of CEE-PMLP was achieved. It can be seen that the overall prediction performance was better, and the model could also predict points with large jumps caused by time series. To quantitatively assess the forecast performance of CEE-PMLP, six forecasting methods were used as comparisons and four evaluation criteria were calculated (Table 3). These comparative models included four single-prediction models (gated recurrent unit (GRU), multilayer perceptron (MLP), random forest (RF), and support vector regression (SVR)) and two hybrid models (particle swarm optimization–multilayer perceptron (PSO-MLP) and empirical mode decomposition–particle swarm optimization–multilayer perceptron (EMD-PSO-MLP)).

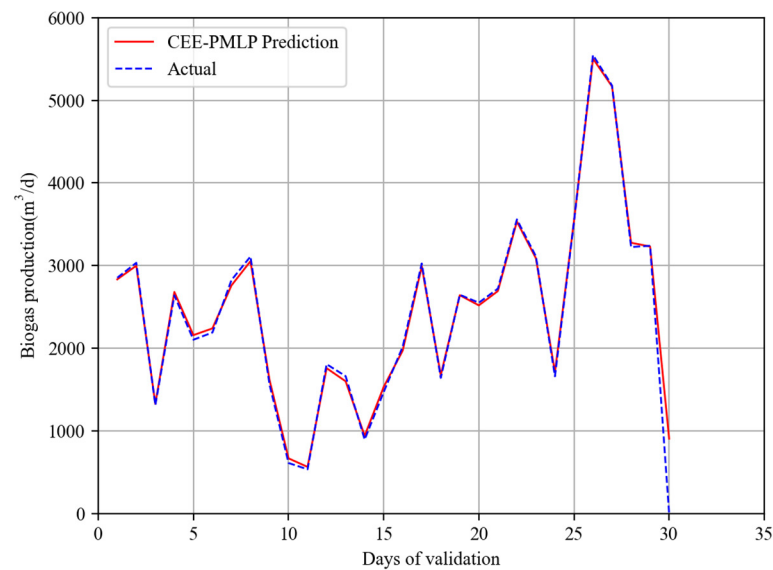


Figure 6. Comparison between experimental results and model simulations over 30 days.

Table 3. Comparison of prediction forecast accuracy between different models.

Model Name	R ²	RMSE	MAPE	MAE
CEE-PMLP	0.98	112.571	5.948	64.25
EMD-PSO-MLP	0.92	140.6	7.21	109.18
PSO-MLP	0.72	200	11	170
MLP	0.65	205	12	175
GRU	0.39	210	12.5	180
SVR	0.38	215	13	185
RF	0.4	207	12.3	178

Compared to the other single models, the MLP demonstrated superior prediction accuracy, with the specific values of R², RMSE, MAPE, and MAE being 0.65, 112.571, 5.948, and 64.25, respectively. The RMSE for MLP was reduced to 205, which was 2.38% to 9.67% lower than that of the other individual models. Similarly, other metrics showed improvements: the MAPE decreased by 2.44% to 7.69%, and the MAE saw a reduction of 1.69% to 5.41%. Although MLP showed some advantages over other models, there was still room for improvement. Thus, MLP was employed as the foundation model when building the hybrid model. With the integration of CEEMDAN and the optimization capabilities of PSO, the CEE-PMLP model outperformed other tested models in terms of prediction accuracy and stability, as shown by its higher R² value (0.98) and lower RMSE, MAPE, and MAE values.

In addition, the complete integration strategy can ensure that the decomposed signal retains all the information of the original signal, further improving the accuracy of the decomposition [32]. Therefore, CEE-PMLP showed the best predictive ability, as can be seen in Table 3. For example, the MAPE value of CEE-PMLP was 5.948, which was decreased by 17.5% compared to EDM-PSO-MLP. The values of RMSE and MAE were also lower than EDM-PSO-MLP, with good predictive ability. At the same time, the R² of CEE-PLMP was 0.98, higher than EDM-PSO-MLP, demonstrating its superior model-fitting and predictive capabilities. Figure 6 shows the predicted curve of the data for the next 30 d. It can be seen that the overall prediction performance was good, and the model was also able to predict points with large jumps caused by time series.

Overall, the CEE-PMLP model proposed in this study successfully predicts biogas production under complex variables. This model effectively smooths the raw data's unevenness caused by the complexity of fermentation process, reduces the risk of overfitting due

to small data volumes, and accurately forecasts possible peaks and valleys, demonstrating significant application value.

Applicability and Generalization:

To evaluate the broader applicability of the proposed CEE-PMLP model, its performance under varying anaerobic digestion (AD) conditions and different feedstocks (substrates) was considered. The model was developed and validated using data from a specific AD system, but its potential generalization to other AD configurations requires further discussion.

The CEE-PMLP model, with its input features such as dry matter (DM), organic matter (OGM), pH, and temperature, is designed to be universally applicable to a broad range of anaerobic digestion (AD) systems. This versatility suggests that the model can be generalized to various substrates, including agricultural residues, food waste, and industrial organic by-products.

However, for substrates with distinct properties like lignocellulosic biomass or lipid-rich waste, the model may not fully capture the nonlinearity or interactions present. Future iterations could integrate additional parameters, like the carbon-to-nitrogen ratio or volatile fatty acid profiles, to improve its generalization across diverse substrates.

The CEE-PMLP model's robustness is attributed to the CEEMDAN decomposition step, which effectively manages the nonstationarity of biogas production data, making it resilient under dynamic operational conditions, such as temperature or pH fluctuations.

The model's framework can be adapted for AD systems operating under thermophilic conditions or with advanced pretreatment methods by retraining with system-specific data, highlighting its flexibility across different AD environments.

The PSO algorithm's global optimization capabilities allow the CEE-PMLP model to learn optimal parameters for different AD systems with minimal manual tuning, positioning it as a strong candidate for transfer learning applications. This means that a pretrained model can be fine-tuned for new datasets with minimal additional data, enhancing its cross-system transferability.

Despite the CEE-PMLP model's superior predictive performance, its broader applicability depends on addressing certain limitations:

The model requires high-quality operational data for training, which may not be available in all AD systems. Enhancing performance on limited or noisy datasets through data augmentation techniques or unsupervised feature extraction could improve its generalizability. Additionally, while CEEMDAN and PSO improve prediction accuracy, they introduce computational complexity. Developing lightweight model versions could facilitate real-time monitoring system deployment.

The model's current reliance on high-quality, structured data for training could be expanded through advanced preprocessing methods, robust feature selection, or ensemble learning to increase its resilience to noisy or incomplete datasets. The complexity of the hybrid approach might limit interpretability for stakeholders unfamiliar with machine learning. Incorporating explainable AI techniques could provide clearer insights into the influence of operational parameters on biogas production.

For real-time monitoring and control of anaerobic digesters, the CEE-PMLP framework needs to reduce computational overhead and integrate with industrial automation systems. Exploring lightweight versions of the model or cloud-based deployment strategies could address this purpose.

By addressing these limitations and exploring potential improvements, the CEE-PMLP model can evolve into a more versatile, scalable, and interpretable tool for biogas production prediction, enhancing efficiency and stability in anaerobic digestion systems.

4. Conclusions

The aim of this study was to develop an accurate and reliable prediction model for biogas production in large-scale ADs. In this context, we proposed a novel hybrid CEE-PMLP biogas prediction model. The findings revealed that the CEE-PMLP hybrid model

delivers superior performance, exhibiting the highest levels of accuracy and efficiency, with a remarkable total R^2 value of 0.98. This fit between the measured and predicted values surpasses those of all other tested models. These results indicated that the proposed CEE-PMLP model can effectively predict the biogas production, providing an alternative new method for biogas prediction to ensure the stability of the complex fermentation process.

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